

P8139 - Homework #4: Genetic Association Study

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1. Introduction

This report conducts a genetic association study using the FMS dataset. We select **Pre_DRM_Max** (baseline dominant arm muscle strength) as the continuous trait and **Dom.Arm** (dominant hand: R vs L) as the binary trait. We perform association analyses on candidate genes under additive model, apply Bonferroni and FDR corrections, and control for population stratification using both PCA and self-reported Race.

2. Descriptive Statistics

```
fms_data <- read.csv("FMS_data.csv", na.strings = c("NA", "", " "))
fms_data$Gender <- as.factor(fms_data$Gender)
fms_data$Race <- as.factor(fms_data$Race)
fms_data$Dom.Arm <- as.factor(fms_data$Dom.Arm)

n_tot <- nrow(fms_data)
desc_stats <- data.frame(
  Characteristic = c("Total records", "Age (mean +/- SD)",
                    "Gender (Male / Female / NA)",
                    "Dominant Arm (R / L / NA)",
                    "Pre_DRM_Max (mean +/- SD)"),
  Value = c(
    n_tot,
    sprintf("%.1f +/- %.1f", mean(fms_data$Age, na.rm=TRUE), sd(fms_data$Age, na.rm=TRUE)),
    sprintf("%d / %d / %d",
            sum(fms_data$Gender == "Male", na.rm=TRUE),
            sum(fms_data$Gender == "Female", na.rm=TRUE),
            sum(is.na(fms_data$Gender))),
    sprintf("%d / %d / %d",
            sum(fms_data$Dom.Arm == "R", na.rm=TRUE),
            sum(fms_data$Dom.Arm == "L", na.rm=TRUE),
            sum(is.na(fms_data$Dom.Arm))),
    sprintf("%.2f +/- %.2f",
            mean(fms_data$Pre_DRM_Max, na.rm=TRUE),
            sd(fms_data$Pre_DRM_Max, na.rm=TRUE))
  )
)
kable(desc_stats, caption = "Summary of Participant Characteristics")
```

Table 1: Summary of Participant Characteristics

Characteristic	Value
Total records	1397
Age (mean +/- SD)	23.7 +/- 5.6
Gender (Male / Female / NA)	426 / 607 / 364
Dominant Arm (R / L / NA)	942 / 90 / 365
Pre_DRM_Max (mean +/- SD)	20.36 +/- 8.60

The dataset has 1397 records but substantial missingness in phenotype and genotype columns; regression models below are fit on complete cases only.

3. Population Stratification (PCA)

We encode each SNP as the count of the minor allele (0/1/2) and mean-impute missing values before PCA.

```
# Count minor-allele dosage from 2-letter genotype strings (e.g. "CT")
encode_snp <- function(g) {
  g <- toupper(as.character(g))
  g[!nchar(g) %in% 2] <- NA
  alleles <- unlist(strsplit(g[!is.na(g)], ""))
  if (length(unique(alleles)) < 2) return(rep(0, length(g)))
  minor <- names(sort(table(alleles)))[1]
  # count minor allele per genotype
  sapply(g, function(x) if (is.na(x)) NA else sum(strsplit(x, "")[[1]] == minor))
}

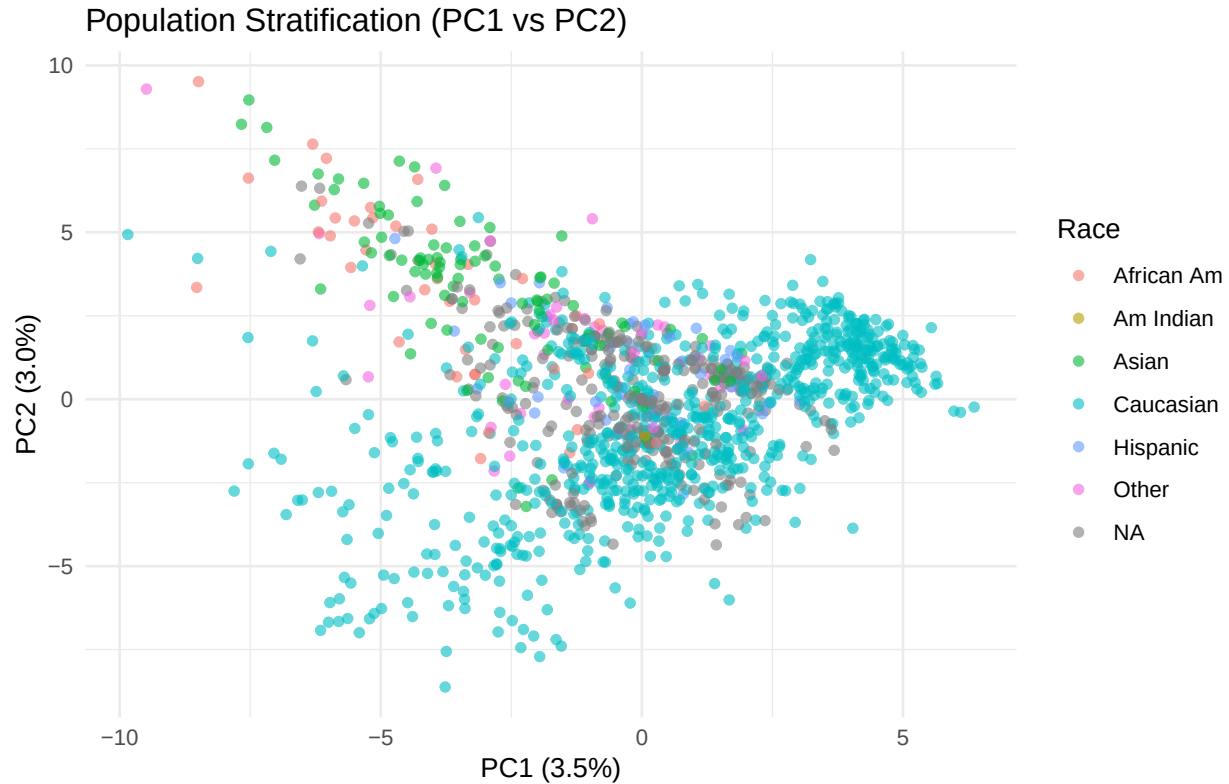
end_col <- which(colnames(fms_data) == "Status")
gene_cols <- colnames(fms_data)[2:(end_col - 1)]

geno_matrix <- sapply(fms_data[, gene_cols], encode_snp)
# mean-impute and drop zero-variance columns
geno_matrix <- apply(geno_matrix, 2, function(c) { c[is.na(c)] <- mean(c, na.rm=TRUE); c })
geno_matrix <- geno_matrix[, apply(geno_matrix, 2, var) > 0]

pca_res <- prcomp(geno_matrix, scale. = TRUE)
ve <- round(100 * summary(pca_res)$importance[2, 1:2], 1)

fms_data$PC1 <- pca_res$x[, 1]
fms_data$PC2 <- pca_res$x[, 2]

ggplot(fms_data, aes(PC1, PC2, color = Race)) +
  geom_point(alpha = 0.6) +
  theme_minimal() +
  labs(title = "Population Stratification (PC1 vs PC2)",
       x = sprintf("PC1 (%.1f%)", ve[1]),
       y = sprintf("PC2 (%.1f%)", ve[2]))
```



PC1 and PC2 jointly explain only 6.5% of genotype variance and the self-reported ancestry groups do not form cleanly separated clusters. Given this limitation, we adopt a pragmatic dual strategy: we still include **PC1 and PC2** as covariates (they absorb whatever coarse structure is detectable), and we **additionally adjust for self-reported Race** as a categorical covariate, which is the more reliable stratification signal in this cohort.

4. Genetic Association Analysis

To avoid selection bias from cherry-picking candidate SNPs, our **primary analysis** is a full-panel scan: every SNP in the FMS panel is tested under a single **additive model** for each trait, adjusting for Age, Gender, PC1, PC2, and self-reported Race. Using one pre-specified model per SNP gives a clean multiple-testing denominator (number of SNPs, not SNPs $\times$ models), so Bonferroni and Benjamini-Hochberg FDR corrections are valid.

4.0 LD structure between SNPs

SNPs within the same gene are often in linkage disequilibrium (LD), which would make a naive Bonferroni correction (dividing by N) overly conservative. Before interpreting any adjusted P value we therefore check how correlated the SNPs in this panel actually are.

```
cor_mat <- cor(geno_matrix, use = "pairwise.complete.obs")
abs_cor <- abs(cor_mat[upper.tri(cor_mat)])
N_snp <- ncol(geno_matrix)

ld_summary <- data.frame(
  Metric = c("Number of SNPs (N)", "Mean |r|", "Median |r|",
    "% pairs with |r| > 0.5"),
```

```

Value = c(N_snp,
  sprintf("%.3f", mean(abs_cor)),
  sprintf("%.3f", median(abs_cor)),
  sprintf("%.1f%%", 100 * mean(abs_cor > 0.5)))
)
kable(ld_summary, caption = "LD structure of the SNP panel")

```

Table 2: LD structure of the SNP panel

Metric	Value
Number of SNPs (N)	215
Mean $ r $	0.037
Median $ r $	0.025
% pairs with $ r > 0.5$	0.5%

Pairwise correlations are very low: mean $|r| \approx 0.037$ and only 0.5% of pairs exceed 0.5. In other words, the 215 SNPs in this panel are close to mutually independent, so **standard Bonferroni correction using N is appropriate here**. We therefore use the usual Bonferroni and Benjamini-Hochberg FDR adjustments in the scans below.

```

fit_snp <- function(gene, trait, model = "Additive", family = "gaussian") {
  x <- encode_snp(fms_data[[gene]])
  if (length(unique(na.omit(x))) < 2) return(NULL)
  if (model == "Dominant") x <- as.integer(x >= 1)
  if (model == "Recessive") x <- as.integer(x == 2)
  if (length(unique(na.omit(x))) < 2) return(NULL)

  df <- data.frame(y = fms_data[[trait]], x = x,
    Age = fms_data$Age, Gender = fms_data$Gender,
    PC1 = fms_data$PC1, PC2 = fms_data$PC2,
    Race = fms_data$Race)

  fit <- tryCatch(
    if (family == "gaussian") lm(y ~ x + Age + Gender + PC1 + PC2 + Race, data = df)
    else glm(y ~ x + Age + Gender + PC1 + PC2 + Race, data = df, family = "binomial"),
    error = function(e) NULL)
  if (is.null(fit) || !"x" %in% rownames(summary(fit)$coefficients)) return(NULL)
  co <- summary(fit)$coefficients["x", ]
  data.frame(Gene = gene, Model = model,
    Estimate = if (family == "gaussian") co[1] else exp(co[1]),
    P = co[4])
}

scan_panel <- function(trait, family, model = "Additive") {
  res <- do.call(rbind, lapply(gene_cols,
    function(g) fit_snp(g, trait, model, family)))

  res$P_Bonf <- p.adjust(res$P, "bonferroni")
  res$P_FDR <- p.adjust(res$P, "fdr")
  res[order(res$P), ]
}

fms_data$DomArmL <- ifelse(fms_data$Dom.Arm == "L", 1,
  ifelse(fms_data$Dom.Arm == "R", 0, NA))

```

4.1 Continuous trait: Pre_DRM_Max (additive scan)

```
scan_cont <- scan_panel("Pre_DRM_Max", "gaussian")
n_cont <- nrow(scan_cont)
colnames(scan_cont)[3] <- "Beta"
kable(head(scan_cont, 10), row.names = FALSE, digits = 4,
       caption = sprintf("Top 10 additive associations with Pre_DRM_Max (full-panel scan, %d SNPs tested)", n_cont))
```

Table 3: Top 10 additive associations with Pre_DRM_Max (full-panel scan, 200 SNPs tested)

Gene	Model	Beta	P	P_Bonf	P_FDR
p2ry2_rs1783596	Additive	0.6594	0.0052	1	0.9835
prdx6_4382766	Additive	-0.7463	0.0148	1	0.9835
lepr_8179183	Additive	1.0154	0.0213	1	0.9835
carp_c105t	Additive	0.5300	0.0381	1	0.9835
pparg_c1a_rs8192678	Additive	-0.5234	0.0438	1	0.9835
akt1_c6024t_c8166t	Additive	1.0128	0.0454	1	0.9835
pgcl_g76039a	Additive	-1.6755	0.0468	1	0.9835
vdr_taq1	Additive	-0.5708	0.0505	1	0.9835
carp_a8470g	Additive	-0.4424	0.0614	1	0.9835
acdc_rs1501299	Additive	0.4888	0.0725	1	0.9835

4.2 Binary trait: Dominant Arm (additive scan)

```
scan_bin <- scan_panel("DomArmL", "binomial")
n_bin <- nrow(scan_bin)
colnames(scan_bin)[3] <- "OR"
kable(head(scan_bin, 10), row.names = FALSE, digits = 4,
       caption = sprintf("Top 10 additive associations with Dom.Arm L vs R (full-panel scan, %d SNPs tested)", n_bin))
```

Table 4: Top 10 additive associations with Dom.Arm L vs R (full-panel scan, 200 SNPs tested)

Gene	Model	OR	P	P_Bonf	P_FDR
tnni1_2799672	Additive	3.3629	0.0075	1	0.9200
nr3c1_rs4582314	Additive	1.6257	0.0114	1	0.9200
p2rx2_6560891	Additive	1.8336	0.0138	1	0.9200
pcr5_snp2	Additive	1.8072	0.0190	1	0.9484
tpd52l1_3778458	Additive	1.4355	0.0350	1	1.0000
igf2_3213220	Additive	4.6172	0.0449	1	1.0000
ppar_gp12a	Additive	0.5592	0.0521	1	1.0000
nr3c1_rs4634384	Additive	1.4380	0.0564	1	1.0000
bcl6_17797517	Additive	1.3919	0.0565	1	1.0000
apoe_c472t	Additive	3.0492	0.0676	1	1.0000

5. Summary and Conclusions

The analytic cohort comprises 1397 records of young adults (age 23.7 +/- 5.6 years), predominantly female (607 female vs 426 male among non-missing) and predominantly right-handed (942 right vs 90 left), with mean baseline dominant-arm strength $\text{Pre_DRM_Max} = 20.36 \pm 8.60$. Self-reported ancestry is overwhelmingly Caucasian with smaller Asian, African American, and Hispanic subgroups, and missingness is substantial across both phenotype and genotype columns, so all regression models are fit on complete cases. After quality filtering, 215 biallelic SNPs entered the full-panel scan.

1. **Muscle strength (continuous).** An additive scan of 200 SNPs adjusted for Age, Gender, PC1, PC2, and Race showed that no SNP passes either Bonferroni or FDR correction, i.e. we report a null result for this trait; the top nominal hits in the table above should be interpreted as hypothesis-generating only. The top nominal signal was **p2ry2_rs1783596** ($\beta = 0.66$, $P = 0.00516$, $P_{\text{Bonf}} = 1$, $P_{\text{FDR}} = 0.984$).
2. **Hand preference (binary).** A logistic scan of 200 SNPs with the same covariates showed that no SNP passes either Bonferroni or FDR correction, i.e. we report a null result for this trait; the top nominal hits in the table above should be interpreted as hypothesis-generating only. The top nominal signal was **tnni1_2799672** ($\text{OR} = 3.36$, $P = 0.00751$, $P_{\text{Bonf}} = 1$, $P_{\text{FDR}} = 0.92$), and statistical power was further limited by the small left-handed group (~90 participants).
3. **Stratification and correction.** PC1 and PC2 jointly explain only 6.5% of genotype variance and do not form clean ancestry clusters, which is an expected limitation of candidate-gene PCA, so self-reported Race was added as the primary ancestry control. Pairwise SNP-SNP LD is very low (mean $|r| \approx 0.037$), so standard Bonferroni and Benjamini-Hochberg FDR over N tests are appropriate here. The null result is not an artifact of over-correction: FDR values are also near 1 for all top hits, and the smallest observed P values sit roughly an order of magnitude above the $0.05/N$ threshold.

In conclusion, within this candidate-gene panel and sample size, no SNP predicts baseline muscle strength or hand preference after proper multiple-testing correction, and the top nominal hits reported above should be regarded as hypothesis-generating signals requiring replication in a larger, genome-wide-genotyped cohort before any causal interpretation.